

Homework 6

Due: Wednesday 4/30 at 11:59pm

This assignment is based on the partial implementation of the game Cartographers that I showed you on Friday. The given code (**hw6given.zip**) contains several files, but the only ones that you'll be directly touching are **Scoring.java** and the file for its test cases, **TestScoring.java**. You can open the project by double clicking on **hw6.code-workspace**.

Scoring.java is for the functions that take a grid of terrain values that the user has constructed and determines how many points it is worth. Each of these functions will take a **Grid** object. You can get the dimensions of the grid it represents with its methods **xdim** and **ydim**. Then you can get the **Terrain** object at a certain location using **get**. The possible values of a **Terrain** are the following: **Terrain.OOB** (out of bounds), **Terrain.EMPTY** (nothing here), **Terrain.TREES**, **Terrain.VILLAGE**, **Terrain.FARM**, **Terrain.WATER**, **Terrain.MONSTER** (not normally considered a type of terrain...), **Terrain.MOUNTAIN**, and **Terrain.MOUNTAINW** (represents a mountain with treasure). The **get** method will always return one of these; if you ask for an out of bounds value, it will return **Terrain.OOB**. (You can use this to eliminate some edge cases in your code.)

The test cases in **TestScoring.java** need to create a **Grid** object on which to call the functions in **Scoring.java**. To do this, you'll want to create a 2D array of integers and pass this to the **Grid** constructor. The encoding for each terrain type is included as a comment.

As a sample of a scoring method and a test case for it, I have provided **numTrees** (returns the number of trees in a grid) in **Scoring.java** and **testNumTrees** in **TestScoring.java**.

Complete the problems below and submit your code by creating a .zip file of the directory of code and submitting it through Google Classroom. Your .zip file should be named based on your name or username to make it unique (so mine might be **bunde.zip**, **davidbunde.zip**, or **dbunde.zip** for example). You are being asked to return all the code, but should only modify **Scoring.java** and **TestScoring.java**.

Here are the problems:

1. (12 points) Within the file **TestScoring.java**, write JUnit test cases for the following functions in the **Scoring** class:
 - (a) **Scoring.seaside**, which takes a **Grid** and returns the number of village terrains it contains that are (horizontally or vertically) adjacent to at least one water terrain.
 - (b) **Scoring.villageSize**, which takes a **Grid** and a location (x and y coordinates as separate integers). It returns the number of village locations reachable from that location only going through village terrain. (If you think of each terrain as part of a larger village, this is the total size of the village.) If the given location does not contain village terrain, the method returns 0.
2. (12 points) Implement each of the functions listed above in the file **Scoring.java**. To implement **villageSize**, you should use the backtracking strategy from Lab 5. (Note that a **Point** class is provided here as well. You can access its coordinates using its attributes **x** and **y** and also create a **HashSet<Point>** containing a collection of **Point** objects.)
3. (1 point) At the top of **TestScoring.java**, put a statement about collaboration.